

CHAPTER 17

In search of a temporal niche: Environmental factors

Roelof A. Hut^{†,*}, Noga Kronfeld-Schor[‡], Vincent van der Vinne[†] and
Horacio De la Iglesia[§]

[†] *Chronobiology unit, Center for Behaviour and Neurosciences, University of Groningen, Groningen, The Netherlands*

[‡] *Department of Zoology, Tel Aviv University, Tel Aviv, Israel*

[§] *Department of Biology and Program of Neurobiology and Behavior, University of Washington, Seattle, USA*

Abstract: Time as an ecological niche variable or “temporal niche” can be defined in the context of the most prominent environmental cycles, including the tidal cycle, the lunar day and month, the solar day, and the earth year. For the current review, we focus on the 24-h domain generated through the earth’s rotation around its axis (solar day). The daily environmental cycles of light and temperature are a dominant ecological factor generating a variety of adaptations among animals. In this review, we describe these adaptations with a special focus on the visual system and on the adaptive plasticity of activity patterns. Our goals are: (1) Underscore the importance of the 24-h time axis as critical variable in the ecological niche. (2) Highlight cases of temporal niche switches at the evolutionary timescale (phylogenetic level). (3) Review temporal niche switching within an individual’s lifespan. (4) Evaluate possible underlying mechanisms for temporal niche switching. (5) Describe a new hypothesis of circadian thermoenergetics which may explain several cases of temporal niche switching in mammals. With this, we hope to inspire experiments under natural conditions or more complex laboratory environments, aimed to reveal environmental factors and mechanisms underlying specific temporal programs.

Keywords: temporal niche switching; circadian thermoenergetics hypothesis; nocturnality; diurnality; eye properties; visual system; environment; evolution.

Introduction

More than half a century after G. Evelyn Hutchinson’s crisp definition of ecological niche as a multidimensional space delineated by the

*Corresponding author.

Tel.: +31503632060; Fax: +31503632148

E-mail: r.a.hut@rug.nl

range of resources in which a species survives and reproduces (Hutchinson, 1957), “time” still struggles to make it into the list of critical niche orthogonal axes (Kronfeld-Schor and Dayan, 2003, 2008). Those of us studying how time shapes physiology and behavior take the inclusion of time in this multidimensional space for granted; after all, it is hard to imagine the survival of Arctic birds hatching just before the winter solstice or honeybees foraging at midnight. Yet, we feel that studies that have addressed which environmental factors and mechanisms shape this temporal niche deserve a closer look.

The daily temporal niche of a species is usually defined as the time of day at which its individuals display locomotor activity. Thus, diurnality and nocturnality are defined as temporal patterns with activity, respectively, during daytime and night time. Similarly, crepuscularity is characterized by activity around dawn and dusk and cathemerality by activity both during the daytime and nighttime. These definitions have been biased to a great extent by the methodology to measure locomotor activity. For instance, measuring running-wheel activity may portray a different locomotor-activity pattern than patterns emerging from infrared-beam interruptions. The presence of a running-wheel *per se* can even cause a shift in temporal distribution of locomotor activity as it is the case in Nile rats (Blanchong et al., 1999), degus (Kas and Edgar, 1999), Mongolian gerbils (Weinert et al., 2007), and golden spiny mice (Table 1; Cohen et al., 2009). Importantly, locomotor activity displayed by an animal under laboratory conditions may—regardless of the measurement method—represent the expression of different behavioral programs under natural conditions (Fig. 1). In other words, wheel running may reflect foraging behavior, predator avoidance, mate searching, or even escape behavior. Thus, it is important to define the conditions under which locomotor activity is measured and the methodology.

Circadian studies under narrow laboratory conditions have led to a dichotomous view of temporal niche in which animals display either

nocturnal or diurnal phenotypes. However, the temporal programs under natural conditions, where environmental factors are more variable and the behavioral repertoire of an animal has a wider range, are likely more complex and more plastic. A classic example of this complexity is the locomotor-activity patterns migratory birds display in the laboratory. Most birds are diurnal and display behaviors such as foraging during the daytime; under laboratory conditions, this is usually manifested by increased locomotor activity during the light phase. During the migratory season, birds migrate during the night; under the proper photoperiod, this migratory activity is manifested in the laboratory as nocturnal restlessness (Bartell and Gwinner, 2005; Coverdill et al., 2008; Gwinner, 1996). This nocturnal-locomotor activity can be distinguished from the activity birds display during the light phase by an experienced observer. However, an automatic locomotor-activity detection system would only portray a bimodal pattern of activity but would provide no information about the respective behaviors each activity about represents.

The temporal distribution of behavior can also shift under different environmental conditions. In fact, more and more examples of “temporal niche switches,” in which individuals of a species can switch from one temporal niche (e.g., nocturnality) to another (e.g., diurnality) are described both in nature and under laboratory conditions (Table 1; Mrosovsky and Hattar, 2005). Similarly, temporal niche switches occurred across phylogenetic branches of closely related groups. Moreover, within many taxa, these switches occur independently in different phylogenetic branches (e.g., Roll et al., 2006), indicating considerable changes in selective pressure on temporal niche phenotypes and consequently considerable adaptive plasticity in this trait.

Temporal niche switching across phylogeny

Evolutionary plasticity is limited, to a certain extent, by phylogenetic constraints. However,

Table 1. Examples of temporal niche switching in different species

Species		TNS	Cause	Con	References
House mouse	<i>Mus musculus</i>	N > D	Natural conditions	F	Daan et al. (2011)
House mouse	<i>Mus musculus</i>	N > D	Reduced food intake	L	Hut et al. (2011)
House mouse	<i>Mus musculus</i>	N > D	Photoreceptor ablation	L	Mrosofsky and Hattar (2005), Doyle et al. (2008)
House mouse	<i>Mus musculus</i>	N > D	Subthreshold light	L	Doyle et al. (2008)
House mouse	<i>Mus domesticus</i>	N > D	Metabolic demand	L	Perrigo (1987, 1990)
Golden spiny mouse	<i>Acomys russatus</i>	N > D	Interspecies competition	F	Shkolnik (1971)
Golden spiny mouse	<i>Acomys russatus</i>	D > N	Lab conditions	F/L	Levy et al. (2007)
Common vole	<i>Microtus arvalis</i>	N > D ^a	Winter	F	Erkinaro (1969)
Common vole	<i>Microtus arvalis</i>	D > C ^a	Spring	F	Lehmann and Sommersberg (1980)
Field vole	<i>Microtus agrestis</i>	N > D ^a	Winter	F	Erkinaro (1969), Bäumler (1975)
Field vole	<i>Microtus agrestis</i>	N > A	Cage size reduction	L	Lehmann (1976)
Montane vole	<i>Microtus montanus</i>	N > D	Short photoperiod	L	Rowsemitt et al. (1982)
Tundra vole	<i>Microtus oeconomus</i>	N > D ^a	Winter	F	Erkinaro (1969)
Syrian hamster	<i>Mesocricetus auratus</i>	N > C, D	Natural conditions	F/L	Gattermann et al. (2008)
Desert hamster	<i>Phodopus roborovskii</i>	N > N ^b	Interspecies competition	F	Scheibler and Wollnik (2011)
Mongolian gerbil	<i>Meriones unguiculatus</i>	D > N	Running wheel	L	Weinert et al. (2007)
Kangaroo rat	<i>Dipodomys spectabilis</i>	N > D	Drought condition	F	Lockard (1978)
Ord's kangaroo rat	<i>Dipodomys ordii</i>	N > D	Drought condition	F	Boal and Giovanni (2007)
Rat	<i>Rattus norvegicus</i>	N > N, D ^c	Lactation	L	Strubbe and Gorissen (1980)
Rat	<i>Rattus norvegicus</i>	N > D	Natural conditions	F/L	Fenn and Macdonald (1995)
Rat	<i>Rattus norvegicus</i>	N > D	Predation	F	Fenn and Macdonald (1995)
Rat	<i>Rattus norvegicus</i>	N > D	Hypothalamic lesion	L	Richter (1978)
Rat	<i>Rattus norvegicus</i> ; pup	A > N	Development	L	Honma and Hiroshige (1977)
N. American beaver	<i>Castor canadensis</i>	N > FR	Winter	F	Potvin and Bovet (1975), Bovet and Oertli (1992)
Rabbit	<i>Oryctolagus cuniculus</i>	N > D	Predation	F	Bakker et al. (2005)
Degu	<i>Octodon degus</i>	D, C > N, C	Lab conditions, running wheel	F/L	Fulk (1976), Bacigalupe et al. (2003), Garcia-Allegue et al. (1999), Kas and Edgar (1999)
Degu	<i>Octodon degus</i>	D > N	High temperature	L	Vivanco et al. (2010a), Hagenauer and Lee (2008), Refinetti (2006, 2008)
Coruro	<i>Spalacopus cyanus</i>	D > N	Lab conditions	F/L	Ocampo-Garcès et al. (2006), Begall et al. (2002)
Tuco tuco	<i>Ctenomys cf. knighti</i>	D > N	Lab conditions	F/L	Tomotani et al. (2011)

(Continued)

Table 1. Examples of temporal niche switching in different species (*Continued*)

Species		TNS	Cause	Con	References
Blind mole rat	<i>Spalax ehrenbergi</i>	D > N	High temperature, summer	F	Kushnirov et al. (1998), Oster et al. (2002)
Reindeer	<i>Rangifer taradus</i>	D > U	Loss of LD cycle	F	Van Oort et al. (2007)
Roe deer	<i>Capreolus capreolus</i>	C, N > C, D	Winter	F	Bubenik (1960) (in Aschoff, 1962; Tscherkowitsch, 1953)
Coyote	<i>Canis latrans</i>	N, D > N ^d	Hunting	F	Kitchen et al. (2000)
Bat-eared fox	<i>Otocyon megalotis</i>	D > N	High temperature	F	Lourens and Nel (1990)
Geoffroy's cats	<i>Leopardus geoffroyi</i>	N > D	Food shortage	F	Pereira (2010)
Owl monkey	<i>Aotus azarai</i>	C > N	Moonlight	F	Fernandez-Duque et al. (2010)
Rhesus macaque	<i>Macaca mulatta</i>	A > D	Development	L	Tscherkowicz (1953) (in Aschoff, 1962)
Human	<i>Homo sapiens</i> ; infant	A > D	Development	F	Kleitman and Engelmann (1953)
Whiskered bat	<i>Myotis mystacinus</i>	N > D	Spring, autumn	F	Nyholm (1965)
Vampire bat	<i>Desmodus rotundus</i>	LN <> EN	Moonlight avoidance	F	Turner (1975)
Leaf nosed bat	<i>Artibeus lituratus</i>	LN <> EN	Moonlight avoidance	F&L	Erkert (1974), Häussler and Erkert (1978)
Fat-tailed dunnart	<i>Sminthopsis crassicaudata</i>	N > D	Winter	F/L	Warnecke et al. (2008), Morton (1995), M.P. Gerkema (personal communication)
Green toad	<i>Bufo viridis</i>	N > D	Rainfall	F	R.A. Hut (personal observation)
Cane toad, metamorphs	<i>Bufo marinus</i>	D, N > D ^e	Cannibalism	L	Pizzatto et al. (2008)
Atlantic salmon	<i>Salmo salar</i>	A > N	Low temperature	F&L	Fraser et al. (1995), Eriksson (1973)
Atlantic salmon	<i>Salmo salar</i>	N > D	Reduced food	L	Metcalfe et al. (1999)
Atlantic salmon	<i>Salmo salar</i>	N > D	Growth demand	L	Metcalfe et al. (1998)
Atlantic salmon	<i>Salmo salar</i>	D > C > N	Winter > summer	L	Eriksson (1973)
Brown trout	<i>Salmo trutta trutta</i>	D > C > N	Winter > summer	L	Eriksson (1973)
Golden-lined rabbit fish	<i>Siganus lineatus</i>	D > N	Shore > reef	F	Fox and Bellwood (2011)
Mediterranean lobster	<i>Nephrops norvegicus</i>	N > D	Deep water (>200 m)	F	Chiesa et al. (2010)
Fruitfly	<i>Drosophila melanogaster</i>	C > D	Short photoperiod, low temperature	L	Dubruille and Emery (2008)
Fruitfly	<i>Drosophila melanogaster</i>	C, D > C, N	Moonlight	L	Kempinger et al. (2009)
Midge	<i>Chironomus thummi</i>	N > D	Low temperature	F&L	Kureck (1979)
Seed-harvesting ants	<i>Pheidole</i> spp.	D > N, C	High temperature	F	Whitford et al. (1981)

TNS, temporal niche switch; Con, condition; F, field study; L, lab study; F/L, comparison between field and lab study; F&L, effect observed in field and lab. Switch in the dominant temporal niche indicated as N, nocturnal; D, diurnal; C, crepuscular; U, ultradian rhythms with no diurnal/nocturnal preference; A, arrhythmic; FR, free running circadian rhythm; LN, late night active; EN, early night active.

^aStrong ultradian organization of activity always in either temporal niche preference.

^bEarlier-nocturnal activity.

^cIncreased-diurnal feeding.

^dReduced-diurnal activity.

^eReduced-nocturnal activity.

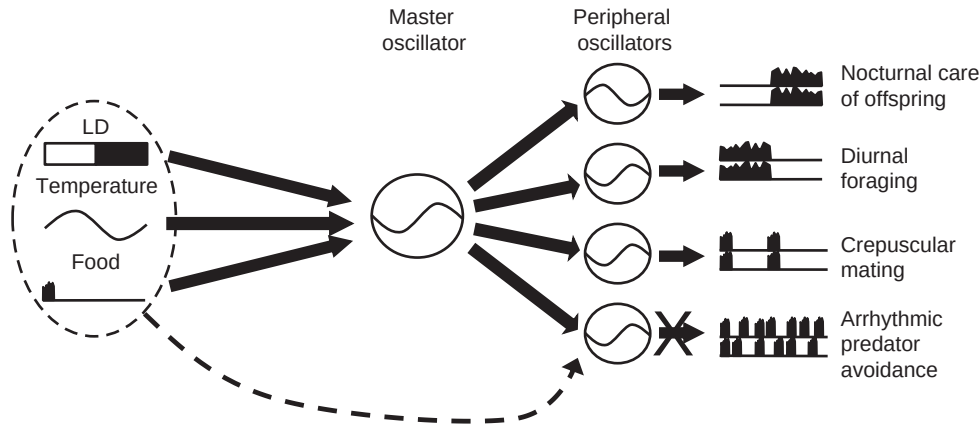


Fig. 1. Complexity of temporal distribution of behaviors. A complex scenario of behavioral programs is likely to be present in natural habitats. Multiple environmental cycles can act as Zeitgebers to entrain a master circadian pacemaker that in turn synchronizes peripheral oscillators. Specific Zeitgebers could also directly entrain peripheral oscillators and bypass the control by the master circadian pacemaker (dotted lines). Peripheral oscillators regulate specific physiological and behavioral outputs that could each have a different temporal niche.

studies focusing on phylogeny as an evolutionary constraint limiting temporal niche switches are scarce. Here, we focus on mammalian taxa for which a phylogenetic framework has been used to map temporal niches.

The mammalian nocturnal bottleneck

The nocturnal bottleneck hypothesis (Crompton et al., 1978; Foster and Menaker, 1993; Heesy and Hall, 2010; Young, 1962) proposes that all mammals share a common nocturnal ancestor. Endothermic mammals evolved from ectothermic reptiles during the Mesozoic between 205 and 65.5 mya. The first mammal-like species are thought to have evolved endothermia in order to occupy the nocturnal temporal niche to avoid ectothermic predators and competitors that were tied to the diurnal temporal niche because of their need for behavioral body temperature regulation through ambient temperature (Crompton et al., 1978). The term “bottleneck” indicates that all mammalian taxa decent from species that faced a common selective pressure favoring nocturnality.

The main arguments in favor of the nocturnal bottleneck hypothesis are threefold: (1) most current mammalian species seem to be nocturnal in their behavior (Roll et al., 2006), (2) mammals have developed endothermia in order to occupy the nocturnal niche, and (3) most mammals have a rod dominated visual system (Table 2) and several cone types (except S and M cones) were lost in mammals, indicating specialization toward a nocturnal life style.

The mass extinction of dinosaurs at the end of the Mesozoic (65.5 mya; Cretaceous–Paleogene extinction event or K/Pg boundary, also called the Cretaceous–Tertiary extinction event or K–T boundary) opened the way for mammalian species to diversify their temporal niches. Diurnality and other forms of temporal niche exploitation emerged through evolutionary radiation and all known temporal niche patterns in mammals were probably present by the end of the Eocene (37 mya) when most modern mammalian taxa had evolved.

Several arguments against the nocturnal bottleneck hypothesis can be raised. First, ectothermia is not exclusively linked to diurnal behavior.

Table 2. Dominant temporal niche and cone photoreceptor percentages

Species		DTN	Cone (%)	References
European ground squirrel	<i>Spermophilus citellus</i>	D	97	Szél and Röhlich (1988)
Mexican ground squirrel	<i>Ictidomys mexicanus</i>	D	95–96	West and Dowling (1975)
13-lined ground squirrel	<i>Ictidomys tridecemlineatus</i>	D	95–96	West and Dowling (1975)
Tupaia	<i>Tupaia belangeri</i>	D	95	Müller and Peichl (1989)
Black-tailed prairie dog	<i>Cynomys ludovicianus</i>	D	90	West and Dowling (1975)
California ground squirrel	<i>Otospermophilus beecheyi</i>	D	85	Kryger et al. (1998)
Eastern gray squirrel	<i>Sciurus carolinensis</i>	D	60	West and Dowling (1975)
Nile grass rat	<i>Arvicanthis niloticus</i>	D, C	35–40	Gaillard et al. (2008)
Degu	<i>Octodon degus</i>	D, N	32	Jacobs et al. (2003)
Plains pocket gopher	<i>Geomys bursarius</i>	N, D	25	Feldman and Phillips (1984), Benedix (1994)
Talas tuco tuco	<i>Ctenomys talarum</i>	D, N	14–31	Schleich et al. (2010)
Magellanic tuco tuco	<i>Ctenomys magellanicus</i>	D, N	10–31	Schleich et al. (2010)
Agouti	<i>Dasyprocta aguti</i>	D, N	10–20	Rocha et al. (2009)
Pig	<i>Sus scrofa domestica</i>	D, N, C	10–20	Hendrickson and Hicks (2002)
Guinea pig	<i>Cavia porcellus</i>	C, A	8–17	Peichl and González-Soriano (1994)
Mongolian gerbil	<i>Meriones unguiculatus</i>	N, D	13	Govardovskii et al. (1992)
European mole	<i>Talpa europaea</i>	A, U	10–12	Glösmann et al. (2008)
Ansell's mole rat	<i>Cryptomys anelli</i>	N, D, A	10	Peichl et al. (2003)
Giant mole rat	<i>Cryptomys mechowii</i>	N, D, A	10	Peichl et al. (2003)
Coruro	<i>Spalacopus cyanus</i>	N, D	10	Peichl et al. (2005)
Common shrew	<i>Sorex araneus</i>	A, U	6–13	Peichl et al. (2000), Crawford (1953)
Greater white-toothed shrew	<i>Crocidura russula</i>	N	5–6	Peichl et al. (2000), Crawford (1953)
Human	<i>Homo sapiens</i>	D	4.7	Curcio et al. (1990)
Fraser's musk shrew	<i>Crocidura poensis</i>	N	4–5	Peichl et al. (2000)
Rabbit	<i>Oryctolagus cuniculus</i>	N, D, C	4	Peichl (1997)
Mouse	<i>Mus musculus</i>	N	3	Carter-Dawson and LaVail (1979)
Lesser hedgehog, tenrec	<i>Echinops telfairi</i>	N	1.5–2.3	Peichl et al. (2000)
Rat	<i>Rattus norvegicus</i>	N	1	LaVail (1976), Szél and Röhlich (1992)
Eastern woodrat	<i>Neotoma floridana</i>	N	1	Feldman and Phillips (1984)
Mouse opossum	<i>Thylamys elegans</i>	N	0.4–1.2	Palacios et al. (2010)

Cones are functional at higher light intensities than rods and therefore thought to be indicative of diurnal vision. Consequently, cone densities, as percentage of the total number of photoreceptors in the whole retina, may indicate the dominant temporal niche of a species. DTN, dominant temporal niche; D, diurnal; N, nocturnal; C, crepuscular; A, arrhythmic; U, ultradian.

Geckos, for instance, are small ectothermic reptiles that hunt for prey during the night. Second, it is debatable whether all dinosaurs were ectotherms. The arguments indicating that dinosaurs during the Mesozoic may have evolved forms of endothermic capacity include the discovery of insulating feather structures, fossils of breeding dinosaurs in arctic regions suggesting heat transfer from adults to offspring, raised body posture, fibrolamellar skeletal bone structure, and the presence of nasal

turbinate structures (Nespolo et al., 2011). Even without endothermia, large dinosaurs may have been able to retain enough body heat to support activity during the night through “inertial homeothermy” or “mass homeothermy.” Third, intraocular and orbital bone sizes indicate that several dinosaur species—including relatively small, feathered predatory Theropoda like Velociraptor—seemed to have had eye shapes that facilitate nocturnal vision (Schmitz and Motani, 2011).

Taken together, it seems probable that the nocturnal bottleneck indeed has occurred in early mammals. Recent evidence, however, seems to indicate that potential predators of these early mammals (like some Therapoda species) may have responded to this novel invention of mammalian nocturnal activity by developing endothermic capacities, enabling these dinosaurs to also exploit the nocturnal niche (as indicated by changes in eye morphology). This leads to the interesting conclusion that, already during the Mesozoicum (i.e., before the great extinction of the dinosaurs), the mammalian nocturnal bottleneck triggered an evolutionary arms race between predator and prey for temporal niche occupation. This in turn may have led to considerable adaptive plasticity in temporal niche usage among the earliest mammals.

Temporal niche switching in mammals

Rodents

A literature survey of activity patterns of 700 rodent species found that activity patterns at the family level are significantly different from the expected random patterns, emphasizing the strong relationship between intrafamily taxonomic affiliation and daily-activity patterns. Large families (Muridae and Sciuridae) analyzed by subfamilies and tribes showed a similar but stronger pattern than that of the family level. The researchers concluded that phylogeny constrains the evolution of activity patterns in rodents and may limit their ability to use the time niche axis for ecological separation (Roll et al., 2006). Nevertheless, there are some striking examples of temporal niche dichotomy within relatively small taxa. For instance, ground squirrels form a species group which is typically exclusively diurnal. European ground squirrels (*Spermophilus citellus*), unlike diurnal birds, are not even active around dawn and dusk, but appear above ground ~3 h after sunrise and retreat below ground ~3 h before sunset (Hut et al., 1999). In stark contrast to this

extreme diurnal phenotype are flying squirrels, an example of strictly nocturnal Sciuridae (DeCoursey, 1960, 1986).

Primates

The temporal niche of primates ranges from diurnality to nocturnality with many species showing “cathemeral” patterns of activity. Although the term cathemeral has been widely accepted in the primatology literature, a careful analysis of so-called cathemeral patterns of activity indicates that these patterns are crepuscular, with varying degrees of nocturnality and diurnality, which is in many cases determined by environmental factors (see *Luminance and Temporal niche switching*). Strepsirrhine primates (lemurs and lorises) are predominantly nocturnal. Haplorhine primates (tarsiers, monkeys, and apes), in contrast, are predominantly diurnal, with two genera representing the nocturnal exception: *Tarsius* (tarsiers) and *Aotus* (owl monkeys). Thus, owl monkeys represent a remarkable exception among anthropoid primates (Old World monkeys, New World monkeys, apes, and humans), as the only nocturnal genus. Because strepsirrhine primates and tarsiers are typically nocturnal and both groups show more primitive features than anthropoid primates, it is traditionally believed that the ancestral Eocene primate was probably nocturnal (Martin and Ross, 2005). This notion, however, has been challenged by molecular evolution studies of the S-opsin, which usually shows relaxed selection in nocturnal species, of several tarsiers and lemurs (Tan et al., 2005). The authors show great variability in the degree of relaxed selection among nocturnal species, suggesting that nocturnality has emerged at different times and multiple times among nocturnal primates, and that the ancestral primate may have been diurnal or cathemeral. Regardless of the temporal niche of the ancestral primate, it is likely that nocturnality–diurnality switches, and diurnality–nocturnality switches have emerged independently and multiple times throughout primate

evolution (Ankel-Simons and Rasmussen, 2008), and this is consistent with the multiple mechanisms that could potentially lead to niche switches.

Temporal niche switching within species

Changes in the daily temporal niche have been reported in a wide range of animals. These changes can be associated with environmental cycles such as seasons and lunar month, with development and aging, and with changing ecological factors such as the presence of competitors, prey, or predators. The golden spiny mouse (*Acomys russatus*) seems to be forced to the diurnal niche under natural conditions, possibly through resource competition with the common spiny mouse (*Acomys cahirinus*). When golden spiny mice are placed under laboratory conditions, they immediately switch back to the nocturnal niche (Fig. 2; Levy et al., 2007). Many such cases of temporal niche switching are briefly summarized in Table 1. Here, we limit our discussion to environmental determinants of niche switches, both in laboratory and in field settings. Social determinants of niche switches, such as inter- and intraspecific interactions are reviewed by Castillo-Ruiz et al. (2012).

Luminance and temporal niche switching

The Mediterranean lobster (*Nephrops norvegicus*) represents a unique case of diurnal–nocturnal temporal niche switching. These animals show

nocturnal emergence from their burrows and foraging during the night when they live in the shallow sea shelf (10–50 m deep), they have a crepuscular phenotype on the lower shelf (50–200 m deep), and they are diurnal in the deeper slope (200–400 m deep; Table 1). Whereas fishermen trolling for this commercially important species had long recognized this depth-dependent temporal niche switch (reviewed in Aguzzi and Sarda, 2008), a recent study has shown that light–dark (LD) cycles that, respectively, mimic the brighter blue light of the shallow waters and the dimmer blue light of the deeper slope are sufficient to determine the nocturnal and diurnal phenotypes (Chiesa et al., 2010). Although similar experiments have not been done in other species with wide ocean-depth ranges, it is likely that the dramatic changes in light intensity and wavelength across water depths represent a key determinant of temporal niche in other species that inhabit the continental margin.

Nocturnal species depending on vision may use moonlight to increase their foraging success. In fact, most nocturnal mammals rely on moonlight, and nocturnal dim light stimulates their locomotor activity. In these species, nocturnal-activity patterns can often be synchronized with moon phases (Erkert, 2008). In the southernmost owl monkey (*Aotus azarai*), this moonlight dependence leads to predictable temporal niche switches with striking nocturnal activity during moonlit nights and crepuscular activity during new moon nights (Table 1, Fernandez-Duque et al., 2010).

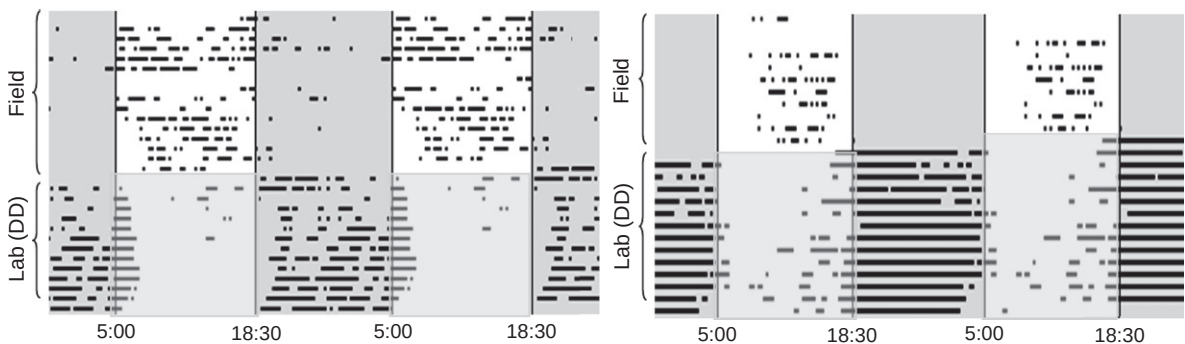


Fig. 2. Temporal niche switching in golden spiny mice: field versus laboratory. Two examples of temporal niche switching in golden spiny mice. When the animals were taken from the field to the laboratory under continuous dim light to assess the phase of their activity rhythm, they show a clear switch from diurnal activity in the field to nocturnal activity in the lab.

In contrast to the stimulatory effects of moonlight on locomotor activity in nocturnal primates and rodents, bats, which rely more on echolocation than on vision, seem to avoid the moonlit part of the night (Erkert, 1974; Häussler and Erkert, 1978; Turner, 1975). This behavior could increase their foraging success by timing hunting when their prey exhibits minimal avoidance behavior and also by reducing the risk of being predated upon.

Ambient temperature and temporal niche switching

In general, temperature differences in the terrestrial environment are much higher than in the buffered aquatic environment. Many ectothermic species may need the high daytime temperatures to augment their body temperature in order to facilitate locomotor activity. In doing so, they face the potential problem of increased evaporative water loss. These effects of temperature can explain why amphibians (with high evaporative water loss through their skin) in general choose the nocturnal niche for terrestrial locomotion, while reptiles (with reduced evaporative water loss through their scaled skin) choose the diurnal niche and even start their active phase with thorough sun basking in the morning. The terrestrial green toad (*Bufo viridis*) escapes daytime high temperatures on the Hungarian puszta plains by hiding in abandoned ground squirrel burrows (Hut and Scharff, 1998). Rainfall allows this animal to emerge above ground in daytime, sometimes in extremely high numbers (Table 1, R.A. Hut, personal observation). A more subtle response to temperature was described in the midge (*Chironomus thummi*), which emerges from the water at dusk to swarm and reproduce at night. Colder water temperatures, however, trigger dawn emergence and daytime swarming. The low water temperature is apparently used by the insects as an indicator for night temperatures being too low to swarm and successfully reproduce (Table 1, Kureck, 1979). The bat-eared fox

(*Otocyon megalotis*) of the South African deserts forages during the night in the summer when mid-day soil temperatures reach 70°C and during the day in the winter when night air temperatures drop to 10°C (Lourens and Nel, 1990). Desert seed-harvesting ants forage during the day in winter but avoid the heat of summer by foraging crepuscularly, nocturnally, or on cloudy days (Whitford et al., 1981). Several rodent species switch activity patterns between seasons, apparently for thermoregulatory reasons. For example, montane voles (*Microtus montanus*) become more diurnal during the winter (Rowsemitt et al., 1982). Similarly, the subterranean mole rat (*Spalax ehrenbergi*) is predominantly diurnal during winter but predominantly nocturnal during summer (Kushnirov et al., 1998; Oster et al., 2002), a behavioral switch which seems to be regulated at the molecular light input pathway of the core clock mechanism (Oster et al., 2002). Also, degus that are maintained at cooler temperatures (e.g., 18 °C) with food *ad libitum*, demonstrate diurnal rhythms in temperature and locomotor activity, whereas degus that are maintained under the same conditions but at more standard animal room temperatures (21–23 °C) shift to more crepuscular or nocturnal patterns (Hagenauer and Lee, 2008). Interestingly, in the field, there is no evidence of degus ever leaving their shelters at night. These data suggest that apart from thermoregulatory costs associated with ambient temperature, animals also integrate other variables like risk of predation, poor nocturnal eyesight, or other variables that may shape activity patterns (Jesseau et al., 2009).

Factors limiting temporal niche switches

Factors limiting temporal niche switches may be divided into internal (the organism anatomy and physiology, such as its sensory and thermoregulatory capabilities) and external (environmental conditions, both biotic and abiotic). As in the case of the factors leading to temporal niche switches,

factors limiting temporal niche switches can be associated with environmental factors, and with inter- and intraspecies interactions. The latter are reviewed by [Castillo-Ruiz et al. \(2012\)](#).

Day and night differ in many environmental factors other than light levels (e.g., temperature, humidity), and therefore expose animals to different challenges that require different anatomical, physiological, and behavioral adaptations. Diurnal animals usually use vision for predatory avoidance and foraging, while nocturnal animals use tactile probing, smell, and hearing. Communication is usually through sound and olfaction in nocturnal animals (and also, although rarely, through bioluminescence). Nocturnal animals may use camouflage (e.g., moths or night jars) and burrowing for concealment from their diurnal predators during the day, whereas diurnal animals on the other hand use visual signals or even conspicuous aposematic coloration ([Daan, 1981](#)). All these complex adaptations to the specific temporal niche may limit temporal niche switches. Here, we focus on some properties of the visual system, the sensory system mostly studied in the context of adaptation to the nocturnal or diurnal temporal niche.

Visual adaptations to specific temporal niches

Night- and day-adapted vision

Because the activity pattern within an individual may show considerable plasticity, it may not be the best parameter to determine the dominant temporal niche phenotype for longer time scale evaluations. Physiological and anatomical determinants for temporal niche usage likely integrate over a much longer time frames (generations) than individual activity patterns do. The best candidate for such a determinant is likely to be found in the eye, since this organ is specifically adapted to light levels that range—in up to five orders of magnitude—between the darkness of the night and the bright light during the day. These visual

adaptations can take place at the anatomical, physiological, or molecular level to define a generally nocturnal or diurnal life style for a specific species.

The adaptive value of a temporal niche will likely present trade-offs due to selective forces that act upon specific physiological and anatomical systems. For instance, in a species forced to become diurnal to escape nocturnal predators new physiological and behavioral strategies may need to be selected to cope with activity at the warmer phase of the day. The most prominent and predictable environmental variable that may exert selective pressure in association with specific temporal niches is the dramatic change in luminance across the 24-h day. Several studies have focused on adaptations of the visual system as key features of evolutionary adaptation to specific temporal niches.

The large majority of reef fish families are primarily diurnal. Nocturnality has been observed in at least 13 families across teleost reef fishes (reviewed in [Schmitz and Wainwright, 2011](#)). An analysis of morphological traits of the eyes in 265 species of teleost reef fish in 43 different families ([Schmitz and Wainwright, 2011](#)) revealed that the eye morphology of nocturnal reef teleosts is characterized by better light sensitivity, resulting from large relative eye size, high optical ratio, and large, rounded pupils. Improved dim-light image formation comes at the cost of reduced depth of focus and reduction of potential accommodative lens movement. Diurnal teleost reef fish have much higher morphological and optical diversity than nocturnal species, with large ranges of optical ratio, depth of focus, and lens accommodation. The authors suggest that the trade-off between improved dim-light vision and reduced optical diversity may be a key factor in explaining the lower trophic diversity of nocturnal reef teleosts. Thus, it is likely that the trade-off associated with nocturnal vision may represent a limiting factor for evolutionary radiation in nocturnal taxa.

Several studies on different taxonomic groups tested the effect of switching from diurnal; vision

based foraging, to nocturnal foraging, showing that low light levels can constrain visual abilities and reduce foraging success. For example, the common murre (*Uria aalge*), a diurnal, diving predator seabird, extends its activity to the night during the breeding season. Diving depths decreased with decreasing nocturnal light levels, probably decreasing foraging efficiency (Regular et al., 2011). Juvenile salmonids are visual foragers; they switch from being predominantly diurnal for most of the year to being nocturnal in winter. Their foraging efficiency is high at light intensities down to those equivalent to dawn or dusk but drops markedly at lower levels of illumination; even under the best night condition (full moon and clear sky), the feeding efficiency is only 35% of their diurnal efficiency (Fraser and Metcalfe, 1997).

In the nocturnal house mouse, the retina is dominated by rods but also contains 3% cones (Table 2). This suggests that the mouse retina retained diurnal visual function. Indeed, at the functional level, Cameron et al. (2008) described increased cone functionality in the subjective day and increased rod functionality during the subjective night. This indicates that an endogenous circadian clock prepares the mouse retina for visual function during daytime. The existence of this daytime visual response indicates that phenotypic plasticity for temporal niche switching on the behavioral level might have generated enough selective advantage to retain some form of diurnal vision in this predominantly nocturnal species.

Eye properties: Rod-cone densities, visual acuity, and sensitivity

The extreme diurnal life style in ground squirrels is associated with two properties of their eyes: the rod/cone ratio and the spectral filtering properties of the lens. The European ground squirrel retina contains virtually only cone photoreceptors (97%; Szél and Röhlich, 1988). Such extreme rod/cone ratios are only found in diurnal mammals,

but the reverse conclusion that lower cone percentages are always associated with nocturnality, cannot be drawn. Much lower retinal cone percentages can be found in other species which also display a diurnal preference, and indeed, they may approach, or even overlap with, cone percentages found in nocturnal species (~10%, Table 2). Some of these diurnal species with relative low cone percentages have a clear overrepresentation of cones in the central retina, either in an elongated region called the visual streak or in a central area called the macula. These areas are tuned to high acuity cone vision and can be found in mammals, birds, amphibians, and reptiles (Rodieck, 1973). Within the macula, some species have a small indent or pit, packed with cone photoreceptors to further increase the acuity in a small central area of the visual field. This structure is called the fovea and can be found in primates (including humans) and birds, which may have two or even three foveae. High density cone areas like the macula and the fovea can be seen as an adaptation to the diurnal niche. In fact, most nocturnal haplorhines (*Tarsius* and *Aotus*), which likely share a diurnal ancestor with diurnal haplorhines, have a fovea; although in *Aotus* individuals, it may be absent (Martin and Ross, 2005).

A study comparing the retinal structure and composition of golden spiny mice (*A. russatus*), which are diurnal at their natural habitat when this is shared with the nocturnal common spiny mice (*A. cahirinus*), found that photoreceptor cells of both species were uniform in shape and tightly packed, and they conformed mainly to the morphology of rods (Kronfeld-Schor et al., 2001). This could indicate that no obvious anatomical adaptation to the diurnal life style has been found in the retina of this species so far. Accordingly, golden spiny mice prefer to forage under boulders, where light intensities are low (22–47 lux; Kronfeld-Schor et al., 2001). These results suggest that the response of the golden spiny mice to become diurnal is either due to the phenotypic plasticity of this species or

that the diurnal life style is a relatively recent evolutionary adaptation.

The increased visual acuity conveyed by structures like the macula and fovea would obviously be advantageous for nocturnal species as well. However, these structures require a reduced pupil diameter to restrict the incoming images from the paraxial regions, and a reduced pupil diameter is not something nocturnal animals can afford as it would reduce the light-gathering ability of the eye (Heesy and Hall, 2010). The way in which nocturnal animals increase visual acuity is by increasing the visual field overlap between eyes (reviewed in Martin and Ross, 2005; Heesy and Hall, 2010). This not only conveys higher acuity but also increases luminance detection because each point in the binocular visual field will gather twice as many photons as a point in the monocular field.

The presence of a tapetum lucidum is seen as an important adaptation to nocturnal life. The tapetum is a membrane located behind the retina that reflects back the light that has gone through the retinal cell layers, increasing light detection by photoreceptor cells. A tapetum is typically present in nocturnal species, such as most strepsirrhine primates. Interestingly, there is a wide variation in the anatomy and histology of the tapetum, indicating that it is an analogous organ that has evolved multiple times from widely different cell types in the eye (Tan et al., 2005).

Eye properties: Spectral filtering and UV protection

Eyes of diurnal species are under permanent photooxidative stress from UV radiation (Ringvold et al., 2000a). It is thought that various levels of yellowish coloration of the mammalian eye lens shields the retina from the (DNA) damaging effects of short-wavelength light (ultraviolet and blue-violet; below ~ 450 nm). Lenses of different species may exhibit different amounts of coloration resulting in different spectral filtering properties. The coloration occurs because of the

interaction of crystallines with a UV filter compound, 3-hydroxykynurenine glucoside. Hence, the mammalian eye lens could be viewed as a low-pass filter with a species specific cutoff value that relates to the species specific life style in the nocturnal–diurnal gradient.

Ground squirrel eyes typically have lenses that appear deep yellow to the human eye, indicating stronger filtering properties in the blue region of human visible light spectrum. Indeed, short-wavelength blue light that is not absorbed by the human lens (cutoff 445 nm) is shown to be absorbed by the ground squirrel lens (cutoff 493 nm; Hut et al., 2000). In contrast, nocturnal animals like the Syrian hamster have a cutoff value ~ 340 nm, allowing UV-A light penetration to the retina (Table 3). The lens filtering property in the 300–500 nm spectral band seems to reflect the long-term dominant temporal niche usage of a species (Fig. 3). This lens filtering property is relatively easy to measure; it has a low within species variance and is easier to interpret than complex retinal properties like photoreceptor distributions (Table 3; Fig. 3). As such, it can be a useful tool for the evaluation of temporal niche switching within individuals.

Eye properties: Ascorbic acid concentration

Another possible mechanism for protecting the eye from UV radiation damages is high concentrations of ascorbic acid in the aqueous humor (Ringvold, 1980). High concentrations of ascorbic acid in the aqueous humor of the eye—20–40 times higher than in the plasma—was reported in many diurnal species including humans, antelopes, tree shrews, golden spiny mice, and rhesus monkeys. Conversely nocturnal mammals, including the slow loris, fruit bats, cats, common spiny mice, and owl monkeys have a concentration similar to that in plasma (Koskela et al., 1989; Reiss et al., 1986; Ringvold, 1980). Interestingly, such a trend was absent in nocturnal and diurnal birds, which apparently use a different mechanism for

Table 3. Filtering properties of the eye lens relates to the species specific temporal niche

Species		DTN	Lens cutoff (nm)	References
European ground squirrel	<i>Spermophilus citellus</i>	D	493	Hut et al. (2000)
Black-tailed prairie dog	<i>Cynomys ludovicianus</i>	D	482	Yolton et al. (1974)
Mexican ground squirrel	<i>Ictidomys mexicanus</i>	D	478	Jacobs and Yolton (1971), Yolton et al. (1974)
Golden mantled ground squirrel	<i>Callospermophilus lateralis</i>	D	471	Yolton et al. (1974)
13-lined ground squirrel	<i>Ictidomys tridecemlineatus</i>	D	470	Chou and Cullen (1984), Yolton et al. (1974)
California ground squirrel	<i>Spermophilus beecheyi</i>	D	464	Jacobs et al. (2003)
Agouti	<i>Dasyprocta aguti</i>	D, N	460	Peichl (1997)
Western gray squirrel	<i>Sciurus griseus</i>	D	447	Yolton et al. (1974)
Human (32years)	<i>Homo sapiens</i>	D	445	Stockman et al. (1999)
Guinea pig	<i>Cavia porcellus</i>	C, A	410	Zigman (1983)
Degu	<i>Octodon degus</i>	D, N	378	Jacobs et al. (2003)
Talas tuco tuco	<i>Ctenomys talarum</i>	D, N	370	Schleich et al. (2010)
Rat	<i>Rattus norvegicus</i>	N	343	Jacobs et al. (2003)
Syrian hamster	<i>Mesocricetus auratus</i>	N	341	Hut et al. (2000), Brainard et al. (1994)
Coruro	<i>Spalacopus cyanus</i>	N, D	330	Peichl et al. (2005)
Magellanic tuco tuco	<i>Ctenomys magellanicus</i>	D, N	325	Schleich et al. (2010)
European mole	<i>Talpa europaea</i>	A, U	310	Glösmann et al. (2008)

The eye lens absorbs short-wavelength light below a certain wavelength in the UV/blue band to protect the retina from tissue damaging properties of high-energy short-wavelength light present during daytime. The wavelength at which transmission is reduced to 50% of the maximal transmittance is presented as lens cutoff value (in nm). DTN, dominant temporal niche; D, diurnal; N, nocturnal; C, crepuscular; A, arrhythmic; U, ultradian.

UV protection. The end product of nitrogen metabolism in birds is uric acid, which similarly to ascorbic acid, is a powerful UV radiation absorber. Mammals, and especially diurnal species, have high concentration of ascorbic acid and low concentration of uric acid in their aqueous, while birds have low concentration of ascorbic acid and high concentration of uric acid in their aqueous (Ringvold et al., 2000b). This suggests that uric acid in birds may have the same UV protection function as ascorbic acid in mammals, and it will be interesting to compare uric acid concentrations in the aqueous humor of diurnal and nocturnal bird species.

Mechanisms underlying temporal niche switching

The timing of specific behaviors throughout the day is the result of the interaction between the

output of circadian clock(s) that are entrained to specific environmental cycles and masking, namely, the direct inhibition or stimulation of behavior by environmental factors. For example, a circadian clock that sustains locomotor activity during the night combined with an inhibitory effect of light on activity will result in a robust nocturnal niche. Three basic mechanisms could result in switching of temporal niche usage, both across phylogeny or within a species or individual (Fig. 4). First, changes in circadian oscillator properties could lead to changes in its phase angle of entrainment, and therefore in the phase of the rhythms, it regulates (see below). Second, a circadian oscillator with essentially the same properties could lead to opposite temporal niches if a switch downstream from the oscillator translates the oscillator output into divergent timing of activity. This seems to be case for the switch between nocturnality and diurnality in rodents. Both, diurnal rats

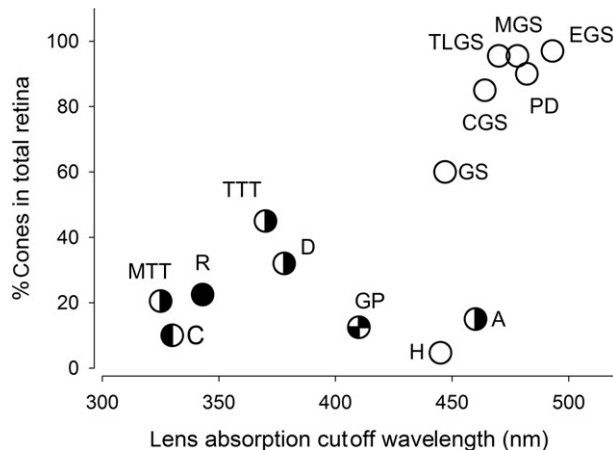


Fig. 3. Cone percentages and lens filtering property relate to dominant temporal niche. Percentage of cones of total number of photoreceptors in the total retina plotted against lens filtering cutoff wavelength (in nm) correlate with each other such that diurnal animals are positioned in the top right corner (with the exception of humans) and nocturnal animals in the bottom left corner. Data taken from [Tables 2 and 3](#), symbol fills correspond to dominant species specific temporal niche. EGS, European ground squirrel; PD, prairie dog; MGS, Mexican ground squirrel; TLGS, 13-lined ground squirrel; CGS, California ground squirrel; A, agouti; GS, gray squirrel (Western and Eastern); H, human; GP, guinea pig; D, degu; TTT, Talas tuco tuco; R, rat; C, cururo; MTT, Magellanic tuco tuco.

(*Arvicanthus* sp.) and nocturnal laboratory rats seem to carry a similarly phased master circadian pacemaker within the suprachiasmatic nucleus (SCN) of the hypothalamus, but the rhythms of locomotor activity and corticosteroid release that this oscillator regulates are oppositely phased ([Kalsbeek et al., 2008](#); [Smale et al., 2003](#)). In accord, the anatomy and the physiology of the SCN in the diurnal and nocturnal golden and common spiny mice show no differences ([Cohen et al., 2010a,b](#)), and both, as well as the degu circadian system ([Hagenauer and Lee, 2008](#)), are largely consistent with that of nocturnal species. Third, expression of a behavioral program during the daytime or nighttime could be solely determined by masking, regardless of the phase of the oscillator. Even if masking does not represent the sole temporal niche determinant, it usually plays a critical role in shaping behavioral outputs of biological timing systems. In the owl monkey *A. azarai*, nocturnal activity regulated by a circadian oscillator is inhibited by darkness during new moon nights,

leading to a prominent nocturnal niche only during moonlit nights. The negative masking by darkness—or positive making by moonlight—was evident after measuring activity during total moon eclipses during full-moon nights, which leads to dramatic decreases in nocturnal activity ([Fernandez-Duque et al., 2010](#)). Moreover, in order to switch to diurnal activity, animals have to overcome the negative masking effect of light. A study comparing masking response to LD pulses in diurnal golden spiny mice and nocturnal common spiny mice found that whereas the nocturnal common spiny mouse display the expected responses, decreasing activity levels in response to a light pulse and increasing activity level in response to dark pulse, golden spiny mice increased their activity in response to a dark pulse (as expected from a nocturnal species), and show insignificant and highly variable response to a light pulse, indicating that in this diurnally active species the negative masking effect of light was indeed removed ([Cohen et al., 2010b](#)). Interestingly,

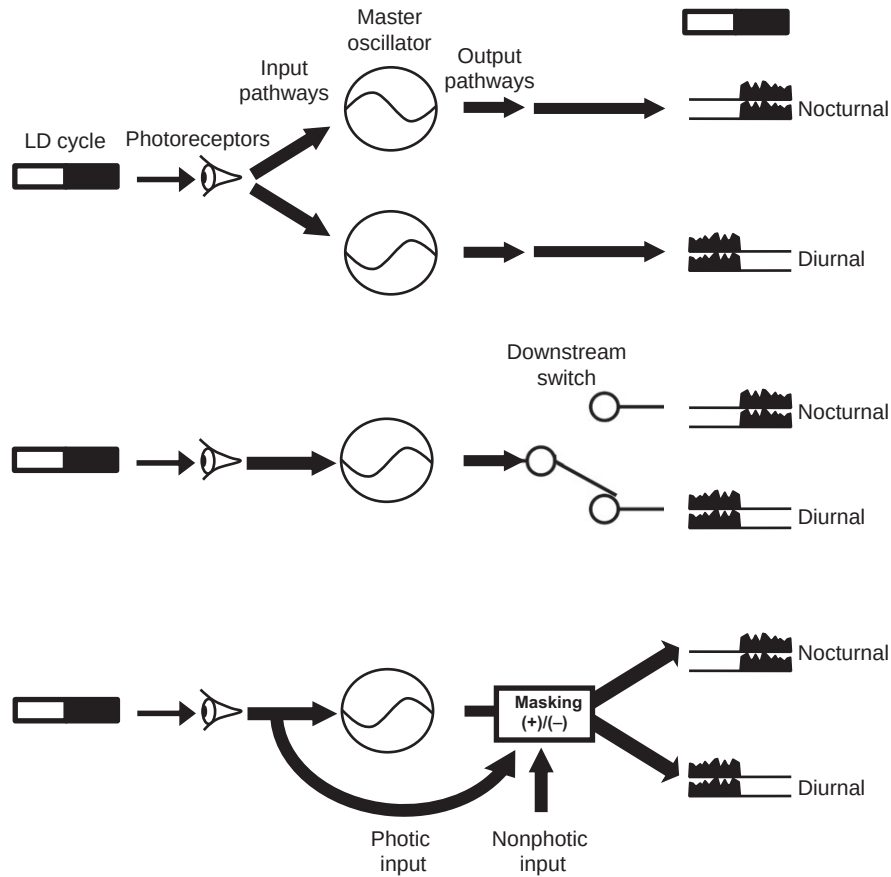


Fig. 4. Mechanisms underlying temporal niche switches in behavioral outputs. Simplified scenario portraying a single behavioral output driven by a circadian oscillator. Top: The properties of the circadian oscillator could change, so that its phase relationship to the LD cycle is different by almost 180° , leading to rhythms of activity that are oppositely phased. Middle: The properties of the oscillator could be unaltered, but a switch in the output pathways regulating locomotor activity could alternate between nocturnal and diurnal patterns of activity. Bottom: The properties of the oscillator could be unaltered, but positive and/or negative masking could determine the rhythmic pattern of activity regardless of the oscillator-phase relationship to the LD cycle. Modified from Chiesa et al. (2010).

diurnal and nocturnal degus present a stable masking by light, each according to its respective chronotype. Thus, whereas diurnal degus increase their activity with light, in nocturnal degus light induce a sharp drop in wheel-running activity (Vivanco et al., 2010b). The resistance to inhibitory effects of light on activity is likely to be essential for a switch into a diurnal niche and possibly occur very early in every evolutionary transition from nocturnality to diurnality.

The top panel of Fig. 4 portrays clocks with different properties leading to differently phased rhythms. Subtle changes in the circadian period of a circadian oscillator can lead to robust changes in the phase angle of entrainment of the oscillator (Daan and Aschoff, 1997; Johnson et al., 2003). In the fruit fly *Drosophila melanogaster*, the temporal organization of locomotor activity is the result of changes in period and phase of their circadian oscillator. The circadian clock that regulates

locomotor activity is constituted by a neuronal network of approximately 150 neurons that express clock genes. The products of these genes constitute the molecular clockwork of biological clocks. *D. melanogaster* flies usually show crepuscular activity. Recent studies have shown that the generation of the morning (M) and evening (E) peaks of locomotor activity are associated with the activity of anatomically identifiable subgroups of cells within the clock neuronal network that oscillate with different phase and period (Grima et al., 2004; Stoleru et al., 2004, 2007; Yoshii et al., 2012). Interestingly, the relative phase between M and E activity is rather plastic, and it is modulated by both room temperature and photoperiod. Shorter photoperiods or lower temperatures will bring the E and M peaks closer together and lead to more consolidated diurnal activity (reviewed in Dubruille and Emery, 2008). The combined effect of photoperiod and ambient temperature likely provides adaptive value; it will push activity to the warmer hours during the winter but to the cooler dawn and dusk during the summer. Remarkably, the shift in the fly's temporal niche relies to some extent on the transcriptional machinery of the molecular clockwork. The clock gene *period* (*per*) has two mRNA isoforms. Under lower temperatures or short photoperiods intron 8 of the gene is more efficiently spliced, and this leads to an earlier appearance of the PER protein, and of the E locomotor-activity peak (Majercak et al., 1999). This effect is further enhanced by the fact that splicing of the *per* gene has a peak during the day. This splicing occurs earlier under short photoperiods, but it is inhibited—and pushed to later times of the day—if temperature is high (Majercak et al., 2004). The effect of photoperiod on *per* mRNA maturation and timing of locomotor activity is synergized by the photodegradation of TIM, the product of the clock gene *timeless* (*tim*). TIM dimerizes with PER and by doing so protects PER from degradation. Under longer photoperiods, because of the photic degradation of TIM, it is hypothesized that it will take longer

for PER to accumulate and for activity to take place. Other mechanisms contribute to the effects of photoperiod and ambient temperature on the timing of expression of molecular clock components. Finally, the fact that the E and M locomotor-activity components rely on different clock neurons offers further plasticity in the fly's clock to independently regulate dawn- and dusk-associated behaviors and their timing (Dubruille and Emery, 2008).

The above scenarios only take into account a master circadian oscillator and a single environmental factor entraining it. The layout will obviously be more complex in natural environments where multiple Zeitgebers will be present (Fig. 1). Further, the presence of multiple peripheral oscillators has been shown in all the species studied so far. The phase of these “slave” oscillators is typically set by a master circadian oscillator, but they can escape the master control. In mammals, specific peripheral oscillators can be entrained by environmental cycles other than light and in an independent manner from the photic entrainment of the master circadian oscillator within SCN (Stokkan et al., 2001). Thus, the physiological and behavioral temporal niche of a species in nature may represent a complex array of rhythms that emerge from a network of interacting circadian oscillators—within and outside the central nervous system—and their respective interactions with several Zeitgebers with different abilities to entrain them.

Adaptive value of temporal niche switching: Unification through the hypothesis of circadian thermoenergetics?

Temporal niche switching examples have been described in many species and under various circumstances (Table 1). The original descriptions are often accompanied by explanations involving the environmental cause which triggers the switch (*proximate* mechanism) or the adaptive value of the switching response (*ultimate* mechanism).

The proximate mechanisms may involve light, temperature, or social interactions. Such mechanisms are discussed in this chapter, in the accompanying paper (social interactions; [Castillo-Ruiz et al., 2012](#)) and in the original papers ([Table 1](#)). Studies evaluating the ultimate mechanisms underlying temporal niche switching rely more on parsimonious reasoning than on solid experimental evidence. Here, we would like to propose a new hypothesis of circadian thermoenergetics (CTE), which may unify the adaptive value of several cases of temporal niche switching.

The ultimate mechanism driving temporal niche switching should involve the optimization process in which energy balance, survival, and reproduction are weighed against each other to maximize fitness. In the temperate zone, nocturnal activity is expected to be more costly because ambient temperatures are high during daytime and low during the night. At nighttime, when ambient temperatures are generally below the thermal neutral zone of most mammals, endothermic animals will energetically profit from reduced heat loss through insulation. This insulation is often found during the rest phase when animals retreat into burrows or nests, huddle together, or adopt a typically sleep-associated thermal insulation posture. Moreover, many endotherms neglect homeothermy and use torpor during the rest phase, making it even more energetically profitable to rest during the night. From an energetics point of view, all endothermic animals in the temperate zone would profit from a diurnal life style. However, other selective forces such as daytime predation pressure or competition may have forced endothermic animals to become or remain nocturnal. This nocturnal phenotype can only be sustained when enough food is available to allow for the energetically costly nocturnal life style. If food becomes scarce, nocturnal animals may return to a diurnal life style to save energetic costs of thermoregulation. Indeed, [Daan et al. \(2011\)](#) observed robust periods of diurnal activity in populations of normally nocturnal house mice kept for 2 years in outdoor enclosures. Roughly,

the same amount of food was delivered in these enclosures on a daily basis, but the mice were allowed to reproduce freely. This led to the hypothesis that food shortage might have triggered the observed temporal niche switch at times of high population density. This hypothesis was tested in the lab by letting lab mice work for their food to simulate natural food shortage. This “work-for-food” protocol indeed induced diurnality in house mice, showing that temporal niche switching can be induced by reduced food intake ([Fig. 5, Hut et al., 2011](#)). This phenomenon has been reproduced in several strains of lab mice (CBA/CaJ, C57bl6, CD1; R.A. Hut, personal communication), as well as in rats (S. Daan, personal communication). The results confirm earlier findings by Perrigo who showed that pregnant female house mice and white footed deer mice changed from nocturnal to a diurnal life style when these mice had to combine working for food with the high metabolic demand of lactation ([Perrigo, 1987, 1990](#)). Together, these findings indicate that, rather than reduced food intake *per se*, the negative energy balance triggered the switch from nocturnal activity to diurnal activity.

The CTE hypothesis can explain these findings by assuming a preserved mechanism in mammals, which associates a diurnal-activity pattern with lower energy requirements than a nocturnal-activity pattern. A negative energy balance (through reduced food intake, lactation, etc.) would induce a switch from nocturnal to diurnal-activity patterns, in order to save energy expenditure through reduced heat loss through increased insulation during the rest phase at night when ambient temperatures are low. In a similar vein, the CTE hypothesis would also predict that heat loss reduction through daily torpor would be optimal when it occurs at night time. This indeed has been found in the house mice in which food intake was reduced by letting these mice work for their food. At the end of the protocol, after the mice robustly occupied the diurnal temporal niche and placed their rest phase at (end of) the night, the mice showed various degrees

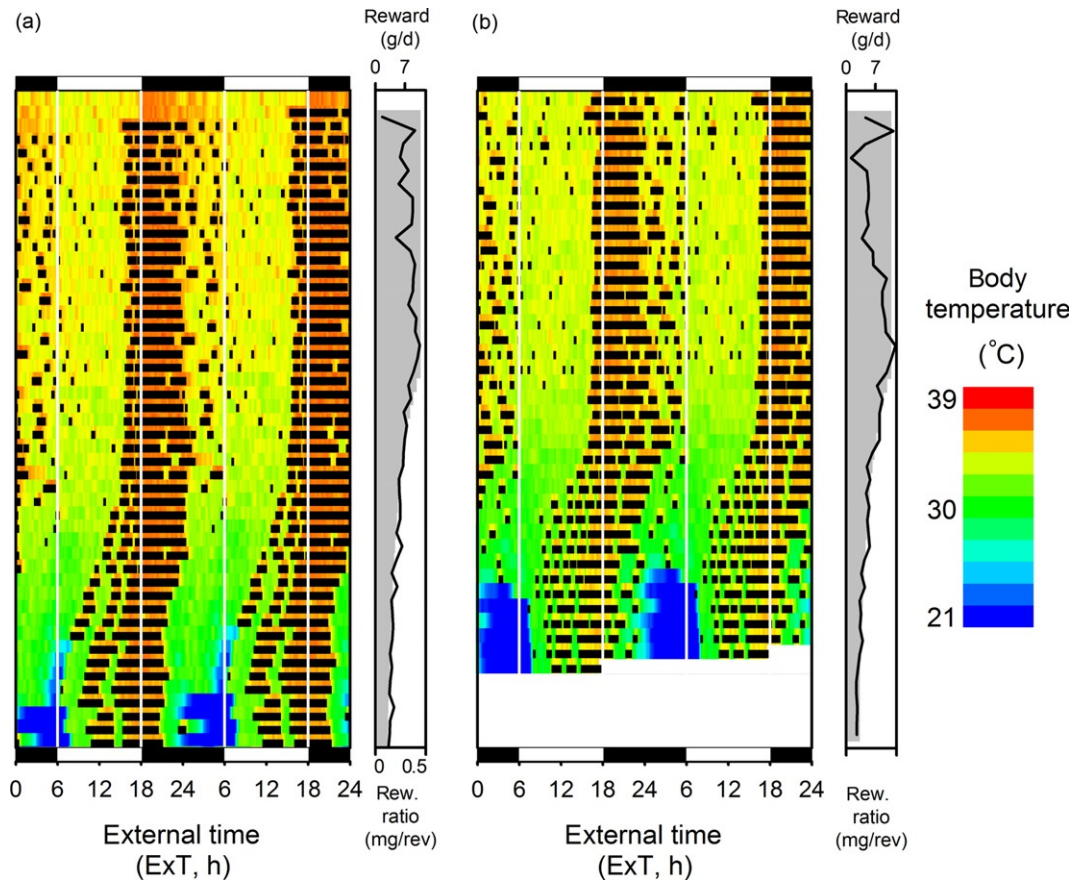


Fig. 5. Experimentally induced temporal niche switching in mice that work for food. Two representative examples (a, b) of mice in which food intake (black line in side graphs) was successfully reduced through decreasing food reward for running-wheel revolutions (gray bars in side graphs). As a result, activity patterns (black bars in actogram) shifted into the light phase of the day (white bars and white lines in actogram). Peak body temperatures (color coded in actograms) also shifted into the day, but body temperatures also gradually lowered over the course of the experiment eventually resulting in clear occurrences of torpor during the rest phase at (end of) the night.

of daily torpor in the dark phase (Fig. 5) (Hut et al., 2011). At this time, environmental temperatures are expected to be the lowest and the energy savings through torpor will be maximized. This nocturnal timing of torpor has also been confirmed in four species of elephant shrew [*Elephantulus rozeti*, *E. myurus* (Lovegrove et al., 2001a,b); *E. rupestris*, *E. edwardi* (Boyles et al., 2012)], in the fat-tailed dunnart (*Sminthopsis crassicaudata*, Warnecke et al., 2008) and in the golden spiny

mouse (*A. russatus*, Levy et al., 2011a,b). It may be that the well studied Djungarian hamster (*Phodopus sungorus*), with daily torpor timed at day-time (Ruby and Zucker, 1992), is an energetically suboptimal exception to this rule.

The fact that the work for food experiments (Hut et al., 2011) and the effect of ambient temperature on circadian rhythms in the degus (Hagenauer and Lee, 2008) mentioned above were carried out under a LD cycle under constant

temperature indicates that the underlying mechanism can use the LD cycle and not temperature cycle *per se* to predict the energetically optimal timing of the activity phase.

The observation that *ad libitum* food supply in the “work-for-food” model (Hut et al., 2011) immediately restores the nocturnal phase under continuous dim light indicates that the circadian pacemaker did not change its phase angle relative to the LD cycle. This leads to the interpretation that the mechanism triggering behavioral temporal niche switching is downstream from the circadian pacemaker (SCN). This situation is schematized in the middle panel of Fig. 4 and is indeed identical to the temporal niche switch in diurnal mammals compared to nocturnal mammals, where activity patterns are in antiphase, but the SCN entrains with a similar phase angle relative to the LD cycle (reviewed in Challet, 2007).

Conclusions

Although temporal niche switching is a complex phenomenon with a manifold of physiological and neurobiological mechanisms and evolutionary causalities, we may be able to draw a number of commonalities. This can be based upon the observation that the main environmental differences between night and day, which are of direct ecological importance for animals living in the wild, relate to temperature and/or light.

Adaptations to the daytime as well as to the nighttime niche are commonly apparent in the eye. Mammals with robust nocturnality like *microchiroptera* bats have strongly reduced vision reflected in their tiny eye size, whereas exclusively diurnal ground squirrels have well developed vision reflected in their large eyes and cone dominated retinas (Table 2; Fig. 3). In other species, long-term adaptations to a specific temporal niche are also present at the level of the retina (e.g., rod and cone densities, Table 2), but sometimes difficult to interpret due to variation in local retinal density (i.e., fovea, visual streak, photoreceptor gradients).

Protection of the retina against high energy short-wavelength light present during daytime is strongly reflected in the filtering properties of the lens (Table 3). For most mammalian species, the lens filtering property together with cone densities (and possibly distribution; Fig. 3) offers a useful measure for assessment of a species' long-term adaptation to the day- or nighttime niche.

Nocturnal animals need low light levels at night to accommodate vision. The interaction with the lunar cycle and the light properties of the environment can explain several cases of temporal niche switching. Such cases have been described above for nocturnal primates and lobsters, but similar responses may be wide spread among nocturnal animals.

Temperature can exhibit important and predictable changes between night and day. When temperatures are too high or too low, animals may modify their activity patterns to reach their optimal body temperature (in both ectotherms and endotherms) or to minimize the energetic needs to regulate their body temperature (in endotherms only). As such, temperature patterns can explain several cases of modification of activity patterns and temporal niche switching. In most regions on earth, the environmental temperature stays well below the lower critical temperatures of most endotherms for a large part of the day. This means that mostly energy is required to maintain euthermic body temperatures. Energy-saving strategies during the rest phase will cause nocturnal activity to be energetically more costly than diurnal-activity patterns. This feature has led us to propose the CTE hypothesis which is a useful tool to understand cases of temporal niche switching induced by temperature, season, or food abundance (Table 1).

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